

The Chinese University of Hong Kong, Shenzhen
School of Science and Engineering

General Chemistry CHM1001
Mid-Term Examination

Date: Sunday, 31 Oct 2021, Time: 15:00 – 17:00

Name: _____ Student ID _____

Answer ALL questions. (Total 14 pages)

Section A: Select the best answer from A-E in each question and put it in the answer sheet provided.

1) The $n = 5$ to $n = 3$ transition in the Bohr hydrogen atom corresponds to the _____ of a photon with a wavelength of _____ nm.

A) absorption, 657

B) absorption, 1280

C) emission, 657

☒ D) emission, 1280

E) emission, 389

2) What is the de Broglie wavelength (m) of a 25-g object moving at a speed of 5.0 m/s?

A) 1.9×10^{32}

☒ B) 5.3×10^{-33}

C) 6.6×10^{-36}

D) 3.32×10^{-36}

E) 3.02×10^{45}

3) An electron cannot have the quantum numbers $n =$ _____, $l =$ _____, $m_l =$ _____.

A) 2, 0, 0

B) 2, 1, -1

C) 3, 1, -1

D) 1, 1, 1

E) 3, 2, 1

4) Which quantum numbers must be the same for the orbitals that they designate to be degenerate in a one-electron system (such as hydrogen)?

A) n , l , and m_l

B) n and l only

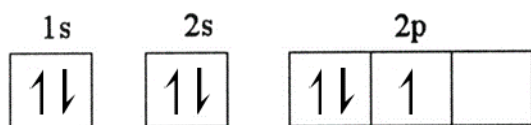
C) l and m_l

D) m_l only

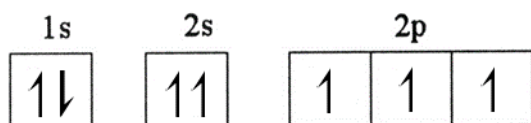
E) n only

5) Which one of the following is the correct electron configuration for a ground-state nitrogen atom?

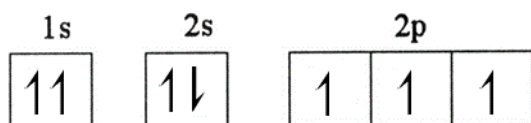
A)



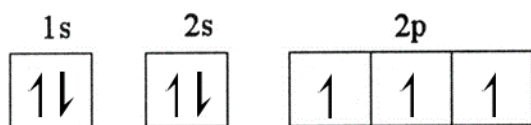
B)



C)



D)



E) None of the above is correct.

6) The ground state configuration of fluorine is _____.

A) $[\text{He}]2s^22p^2$

B) $[\text{He}]2s^22p^3$

C) $[\text{He}]2s^22p^4$

D) $[\text{He}]2s^22p^5$

E) $[\text{He}]2s^22p^6$

7) The effective nuclear charge of an atom is primarily affected by _____.

- A) inner electrons
- B) outer electrons
- C) orbital radial probability
- D) electron distribution
- E) None of the above is correct

8) Which of the following correctly lists the five atoms in order of increasing size (smallest to largest)?

- A) $F < K < Ge < Br < Rb$
- B) $F < Br < Ge < K < Rb$
- C) $F < Ge < Br < K < Rb$
- D) $F < K < Br < Ge < Rb$
- E) $F < Br < Ge < Rb < K$

Consider the following electron configurations to answer the questions 9-10 that follow:

(i) $1s^2 2s^2 2p^6 3s^1$ N_2

(ii) $1s^2 2s^2 2p^6 3s^2$ Mg

(iii) $1s^2 2s^2 2p^6 3s^2 3p^1$ Al

(iv) $1s^2 2s^2 2p^6 3s^2 3p^4$ S

(v) $1s^2 2s^2 2p^6 3s^2 3p^5$ Cl

9) The electron configuration that belongs to the atom with the lowest second ionization energy is _____.

- A) (i)
- ☒ B) (ii)
- C) (iii)
- D) (iv)
- E) (v)

10) The electron configuration of the atom that is expected to have a positive electron affinity is _____.

- A) (i)
- ☒ B) (ii)
- C) (iii)
- D) (iv)
- E) (v)

11) In general, as you go across a period in the periodic table from left to right:

- (1) the atomic radius _____;
- (2) the electron affinity becomes _____ negative; and
- (3) the first ionization energy _____.

- A) decreases, decreasingly, increases
- B) increases, increasingly, decreases
- C) increases, increasingly, increases
- ☒ D) decreases, increasingly, increases
- E) decreases, increasingly, decreases

12) Between which two elements is the difference in metallic character the greatest?

- A) Rb and O
- B) O and I
- C) Rb and I
- D) Li and O
- E) Li and Rb

13) Using the Born-Haber cycle, the ΔH_f° of KBr is equal to _____.

- A) $\Delta H_f^\circ[\text{K (g)}] + \Delta H_f^\circ[\text{Br (g)}] + I_1(\text{K}) + E(\text{Br}) + \Delta H_{\text{lattice}}$
- B) $\Delta H_f^\circ[\text{K (g)}] - \Delta H_f^\circ[\text{Br (g)}] - I_1(\text{K}) - E(\text{Br}) - \Delta H_{\text{lattice}}$
- C) $\Delta H_f^\circ[\text{K (g)}] - \Delta H_f^\circ[\text{Br (g)}] + I_1(\text{K}) - E(\text{Br}) + \Delta H_{\text{lattice}}$
- D) $\Delta H_f^\circ[\text{K (g)}] + \Delta H_f^\circ[\text{Br (g)}] - I_1 - E(\text{Br}) + \Delta H_{\text{lattice}}$
- E) $\Delta H_f^\circ[\text{K (g)}] + \Delta H_f^\circ[\text{Br (g)}] + I_1(\text{K}) + E(\text{Br}) - \Delta H_{\text{lattice}}$

14) Which of the following has the bonds correctly arranged in order of increasing polarity?

- A) Be-F, Mg-F, N-F, O-F
- B) O-F, N-F, Be-F, Mg-F
- C) O-F, Be-F, Mg-F, N-F
- D) N-F, Be-F, Mg-F, O-F
- E) Mg-F, Be-F, N-F, O-F

15) There are _____ valence electrons in the Lewis structure of

$\text{CH}_3\text{OCH}_2\text{CH}_3$.

- A) 18
- B) 20
- ☒ C) 26
- D) 32
- E) 36

16) Resonance structures differ by _____.

- A) number and placement of electrons
- B) number of electrons only
- C) placement of atoms only
- D) number of atoms only
- ☒ E) placement of electrons only

17) A valid Lewis structure of _____ cannot be drawn without violating the octet rule.

- ☒ A) ClF_3
- B) PCl_3
- C) SO_3
- D) CCl_4
- E) CO_2

18) The electron domain and molecular geometry of BrO_2^- is _____.

- A) tetrahedral, trigonal planar
- B) trigonal planar, trigonal planar
- C) trigonal pyramidal, linear
- ☒ D) tetrahedral, bent
- E) trigonal pyramidal, seesaw

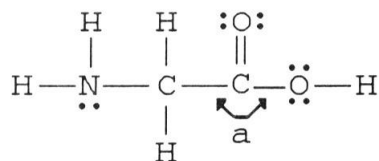
19) The O-C-O bond angle in the CO_3^{2-} ion is approximately _____.

- A) 90°
- B) 109.5°
- ☒ C) 120°
- D) 180°
- E) 60°

20) Of the following species, _____ will have bond angles of 120° .

- A) PH_3
- B) ClF_3
- C) NCl_3
- ☒ D) BCl_3
- E) All of these will have bond angles of 120° .

21) The bond angle marked a in the following molecule is about _____.



- A) 90°
- B) 109.5°
- C) 120°**
- D) 180°
- E) 60°

22) An electron domain consists of _____.

- a) a nonbonding pair of electrons
 - b) a single bond
 - c) a multiple bond
- A) a only
 - B) b only
 - C) c only
 - D) a, b, and c**
 - E) b and c

23) A molecule has the formula AB_3 and the central atom is in a different plane from the surrounding three atoms. Its molecular shape is _____.

- A) tetrahedral
- B) trigonal pyramidal**
- C) linear
- D) T-shaped
- E) bent

24) Of the molecules below, only _____ is polar.

- A) CCl_4
- B) CH_4
- C) SF_4**
- D) SiCl_4

25) Which of the following statements about gases is false?

- A) Gases are highly compressible. ✓
- B) Distances between molecules of gas are very large compared to bond distances within molecules. ✓
- C) Non-reacting gas mixtures are homogeneous. ✓
- D) Gases expand spontaneously to fill the container they are placed in. ✓
- E) All gases are colorless and odorless at room temperature.**

26) Sodium bicarbonate (NaHCO_3) is reacted with concentrated hydrochloric acid (HCl) at 25.0°C and 1.50 atm. The reaction of 7.75 kg of bicarbonate with excess hydrochloric acid under these conditions will produce _____ L of CO_2 .

- A) 1.82×10^3**

B) 2.85×10^4

C) 1.82×10^4

☒ D) 1.50×10^3

E) 8.70×10^2

27) The density of NO_2 in a 4.50 L tank at 760.0 torr and 25.0 °C is _____ g/L.

A) 1.64

B) 9.30

C) 1.68

☒ D) 1.88

E) 3.27

28) The kinetic-molecular theory predicts that pressure rises as the temperature of a gas increases because _____.

A) the average kinetic energy of the gas molecules decreases

B) the gas molecules collide more frequently with the wall

C) the gas molecules collide less frequently with the wall

D) the gas molecules collide more energetically with the wall

☒ E) both the gas molecules collide more frequently with the wall and the gas molecules collide more energetically with the wall

29) As a gaseous element condenses, the atoms become _____ and they have _____ attraction for one another.

A) more separated, more

B) more separated, less

☒ C) closer together, more

D) closer together, less

E) larger, greater

30) In which of the following molecules is hydrogen bonding likely to be the most significant component of the total intermolecular forces?

A) CH_4

B) $\text{C}_5\text{H}_{11}\text{OH}$

C) $\text{C}_6\text{H}_{13}\text{NH}_2$

☒ D) CH_3OH

E) CO_2

31) The intermolecular force(s) responsible for the fact that CH_4 has the lowest boiling point in the set CH_4 , SiH_4 , GeH_4 , SnH_4 is/are _____.

A) hydrogen bonding

B) dipole-dipole interactions

C) London dispersion forces

D) mainly hydrogen bonding but also dipole-dipole interactions

☒ E) mainly London-dispersion forces but also dipole-dipole interactions

32) Hydrogen bonding is a special case of _____.

A) London-dispersion forces

B) ion-dipole attraction

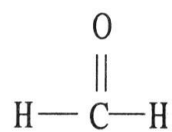
☒ C) dipole-dipole attractions

D) ion-ion interactions

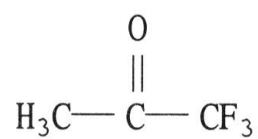
E) none of the above

33) Which one of the following substances will have hydrogen bonding as one of its intermolecular forces?

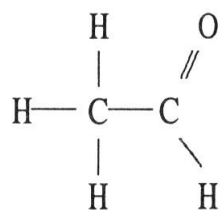
A)



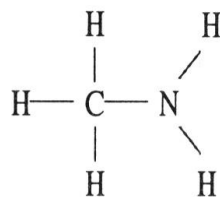
B)



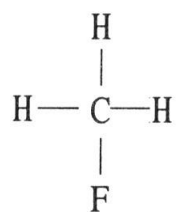
C)



D)



E)



34) Based on the following information, which compound has the strongest intermolecular forces?

Substance	ΔH_{vap} (kJ/mol)
Argon (Ar)	6.3
Benzene (C ₆ H ₆)	31.0
Ethanol (C ₂ H ₅ OH)	39.3
Water (H ₂ O)	40.8
Methane (CH ₄)	9.2

- A) Argon
- B) Benzene
- C) Ethanol
- ☒ D) Water
- E) Methane

35) Which of the following is not a type of solid?

- A) ionic
- B) molecular
- ☒ C) supercritical
- D) metallic
- E) covalent network

36) Crystalline solids _____.

- A) have their particles arranged randomly
- ☒ B) have highly ordered structures

- C) are usually very soft
- D) exist only at high temperatures
- E) exist only at very low temperatures

37) What portion of the volume of each atom or ion on the face of a unit cell is actually within the unit cell?

- ☒ A) $1/2$
- B) $1/4$
- C) $3/4$
- D) all of it
- E) none of it

38) An alloy is a

- A) heterogeneous mixture of two metals.
- B) pure metal.
- ☒ C) metallic material that is composed of two or more elements.
- D) nonmetal with some properties of a metal.
- E) mineral containing two or more metals.

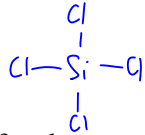
39) NaCl crystallizes in a face-centered cubic cell. What is the total number of ions (Na^+ ions and Cl^- ions) that lie within a unit cell of NaCl?

- A) 2
- ☒ B) 4
- C) 8
- D) 6
- E) 5

40) Inorganic compounds that are semiconductors have an average of _____ valence electrons.

- A) 1
- B) 2
- C) 3
- D) 4**
- E) 5

Section B: Short answer questions. Write your answers in the answer book provided.

1. Consider the compound of SiCl_4 ,
- 
- a) Write the electron configuration for the central atom Si. $[\text{Ne}] 2s^2 2p^2$
 - b) Write the Lewis structure of the molecule.
 - c) Indicate the hybridization of the central atom. sp^3
 - d) Configure the geometry of the molecule. *Tetrahedral*
 - e) Identify the polarity of the molecule. *Non polar*
 - f) CCl_4 is a liquid at room temperature and under normal atmosphere, predict the physical state (gas, liquid, or solid) for SiCl_4 under the same condition.
 - g) What is the dominating intermolecular force between SiCl_4 molecules? *Dispersion*

2. Predict the electron-domain and molecular geometries for (a) SeCl_2 , (b) CO_3^{2-} (c) SnCl_3^-
- Tetrahedral*
Trigonal pyramidal

Tetrahedral
Bent

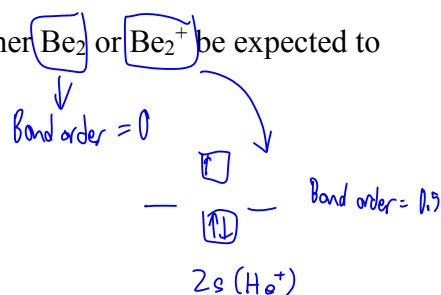
Trigonal planar

3. Predict whether these molecules are polar or nonpolar: (a) BrCl , (b) SO_2 , (c) SF_6 .
- polar*

polar

Nonpolar

4. According to molecular orbital theory, would either Be_2 or Be_2^+ be expected to exist? Explain.



$$\text{Volume \%} = \frac{\frac{4}{3}\pi}{16\sqrt{2}} = \frac{\pi}{3\sqrt{2}} = 74\%$$

5. Calculate the volume percentage of face center cubic lattice occupied by atoms.

$$a = \frac{4r}{\sqrt{2}} \rightarrow \text{Total } V = 16\sqrt{2} r^3, \text{ Atom volume} = \frac{4}{3}\pi r^3 \cdot 4 = \frac{16}{3}\pi r^3$$

6. Which type(s) of intermolecular forces need to be overcome to convert each of the following compounds from liquids to gases?

(a) Methane CH_4 , (b) Methyl fluoride CH_3F , (c) Methyl alcohol CH_3OH

Dispersion

Dipole

-END-

H-bond

Fundamental Constants and equations:

Gas constant: $R=8.3145 \text{ J/mol-K}$

Boltzmann constant: $k= 1.38 \times 10^{-23} \text{ J/K}$

The speed of light (c) is $3.00 \times 10^8 \text{ m/s}$.

Avogadro's number = 6.022×10^{23} molecules

Electronic charge= $1.602 \times 10^{-19} \text{ C}$

$1\text{eV}=1.602 \times 10^{-19}\text{J}$

Rydberg constant $R_H = 1.096776 \times 10^7 \text{ m}^{-1}$

Planck constant $h = 6.62606957 \times 10^{-34} \text{ J-s}$

$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 1.10325 \times 10^5 \text{ Pa} = 101.325 \text{ kPa} = 1.01325 \text{ bar}$

Rydberg Equation:

$$\frac{1}{\lambda} = (R_H) \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

The Bohr Model

$$\Delta E = -hcR_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

The Wave Nature of Matter

$$\lambda = \frac{h}{mv}$$

Einstein's equation

$$E = h\nu = hc/\lambda$$

The Uncertainty Principle

$$(\Delta x) (\Delta mv) \geq \frac{h}{4\pi}$$

ideal-gas equation

$$PV = nRT$$

van der Waals equation:

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

Dipole moment: $\mu = Qr$ (unit: Debyes, $1D = 3.34 \times 10^{-30} \text{ C}\cdot\text{m}$)

Graham's law of effusion:

$$\frac{v_1}{v_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{\rho_2}{\rho_1}} \quad (T, P)$$

Clausius-Clapeyron equation: $\ln P = \frac{-\Delta H_{\text{vap}}}{RT} + C$

Periodic Table of the Elements

Main Group Representative Elements																	
1A 1	2A 2															7A 17	8A 18
1 H 1.00794	2 He 4.002602															3A 13	4A 14
3 Li 6.941	4 Be 9.0121831															5A 15	6A 16
5 Na 22.989770	6 Mg 24.3050	7 Al 26.981538	8 Si 28.0855	9 P 30.973762	10 S 32.065	11 Cl 35.453	12 Ar 39.948									13 B 10.811	14 C 12.0107
19 K 39.0983	20 Ca 40.078	21 Sc 44.955908	22 Ti 47.867	23 V 50.9415	24 Cr 51.9961	25 Mn 54.938044	26 Fe 55.845	27 Co 58.933194	28 Ni 58.6934	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.64	33 As 74.92160	34 Se 78.971	35 Br 79.904	36 Kr 83.80
37 Rb 85.4678	38 Sr 87.62	39 Y 88.90584	40 Zr 91.224	41 Nb 92.90637	42 Mo 95.95	43 Tc [98]	44 Ru 101.07	45 Rh 102.90550	46 Pd 106.42	47 Ag 107.8682	48 Cd 112.414	49 In 114.818	50 Sn 118.710	51 Sb 121.760	52 Te 127.60	53 I 126.90447	54 Xe 131.293
55 Cs 132.905453	56 Ba 137.327	57 La 138.9055	58 Ce 140.116	59 Pr 140.90766	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.964	64 Gd 157.25	65 Tb 158.92534	66 Dy 162.50	67 Ho 164.93033	68 Er 167.259	69 Tm 168.93422	70 Yb 173.04	71 Lu 174.967	72 Hf 178.49
87 Fr [223.02]	88 Ra [226.03]	89 Ac [227.03]	90 Th 232.0377	91 Pa 231.03588	92 U 238.02891	93 Np [237.05]	94 Pu [244.06]	95 Am [243.06]	96 Cm [247.07]	97 Bk [247.07]	98 Cf [251.08]	99 Es [252.08]	100 Fm [257.10]	101 Md [258.10]	102 No [259.10]	103 Lr [262.11]	104 Rf [261.11]
105 Db [262.11]	106 Sg [266.12]	107 Bh [264.12]	108 Hs [269.13]	109 Mt [268.14]	110 Ds [281.15]	111 Rg [272.15]	112 Cn [285]	113 Nh [284]	114 Fl [289.21]	115 Lv [288]	116 Ts [293]	117 Uu [294]	118 Lv [294]	119 Uus [294]	120 Uuh [294]	121 Uub [294]	122 Uut [294]

Lanthanide series	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb
Actinide series	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No

^aThe labels on top (1A, 2A, etc.) are common American usage. The labels below these (1, 2, etc.) are those recommended by the International Union of Pure and Applied Chemistry (IUPAC).
 Except for elements 114 and 116, the names and symbols for elements above 113 have not yet been decided.
 Atomic weights in brackets are the names of the longest-lived or most important isotope of radioactive elements.
 Further information is available at <http://www.webelements.com>
 ** Discovered in 2010, element 117 is currently under review by IUPAC.